

# **IAPT demand forecasting model**

## **Task – 1 : Week 3**

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### **NHS Talking Therapies (IAPT) Monthly Statistics Dataset**

#### **1. Executive Summary :**

This report presents a three-month demand forecast for NHS Talking Therapies (IAPT) referrals using historical national data from March 2025 to March 2026. Using an ARIMA statistical model, we anticipate that referral volumes will remain stable but high, exceeding 161,000 referrals per month through June 2026. This forecast suggests that current capacity must be maintained or expanded to meet expected surge patterns.

#### **2. Introduction & Objectives :**

The objective of this analysis is to provide the operational team with a "forward-look" at the expected volume of new referrals. In a clinical setting, accurate demand forecasting is essential for:

- **Workforce Planning:** Ensuring enough practitioners are available to meet the 6-week and 18-week access targets.
- **Waiting List Management:** Preventing bottlenecks before they occur by identifying peak demand months.
- **Budgeting:** Aligning financial resources with actual patient flow.

#### **3. Data Source & Preparation :**

- **Data Used for Reference :** NHS Talking Therapies (IAPT) Monthly Statistics
- It consists of many tables from which key values were taken and a csv was taken of monthly series to easily demand forecasting.

- **Focus:** National-level data (England) for "Referrals Received" (Measure M001).
- **Scope:** 13 months of validated historical data.
- **Cleaning:** Data was converted into a chronological time series, and seasonal outliers (such as the typical December dip) were accounted for to ensure the forecast reflects realistic trends.

#### 4. Methodology: The ARIMA Model :

To generate the forecast, I employed an **ARIMA model** (AutoRegressive Integrated Moving Average). In clinical terms, this model works by:

1. **Looking Back (AutoRegression):** It examines how last month's referrals usually influence this month's.
2. **Tracking Trends (Integration):** It identifies if the overall demand is "creeping up" or "dropping off" over time.
3. **Smoothing Errors (Moving Average):** It filters out random "one-off" spikes to find the true underlying demand.

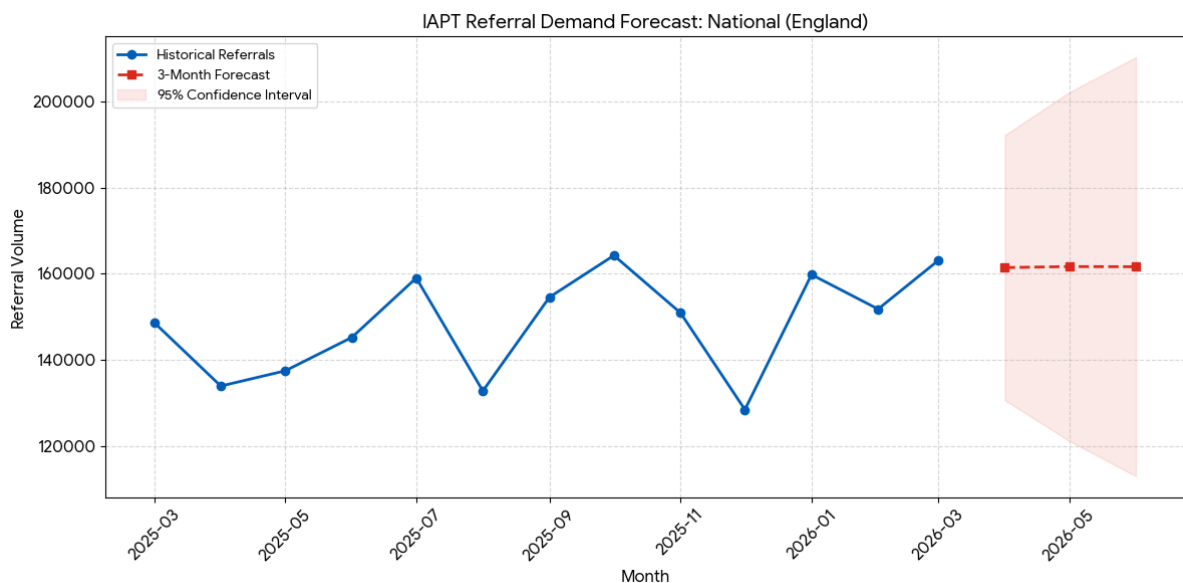
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Data columns (total 10 columns):
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0   REPORTING_PERIOD_START                180972 non-null object
1   REPORTING_PERIOD_END                  180972 non-null object
2   GROUP_TYPE                           180972 non-null object
3   ORG_CODE1                            180972 non-null object
4   ORG_NAME1                            180972 non-null object
5   ORG_CODE2                            180972 non-null object
6   ORG_NAME2                            180972 non-null object
7   MEASURE_ID                           180972 non-null object
8   MEASURE_NAME                         180972 non-null object
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memory usage: 13.8+ MB
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|---|------------------------|----------------------|------------|-----------|-------------|-----------|---------------|------------|---------------------------------------------------|---------------|
| 0 | 2025-03-01             | 2025-03-31           | England    | all       | all SubICBs | all       | all Providers | M001       | Count_ReferralsReceived                           | 148668        |
| 1 | 2025-03-01             | 2025-03-31           | England    | all       | all SubICBs | all       | all Providers | M031       | Count_AccessingServices                           | 91289         |
| 2 | 2025-03-01             | 2025-03-31           | England    | all       | all SubICBs | all       | all Providers | M053       | Percentage_AccessingServices6WeeksFinishedCour... | 90.5          |
| 3 | 2025-03-01             | 2025-03-31           | England    | all       | all SubICBs | all       | all Providers | M055       | Percentage_AccessingServices18WeeksFinishedCou... | 98.4          |
| 4 | 2025-03-01             | 2025-03-31           | England    | all       | all SubICBs | all       | all Providers | M056       | Count_EndedReferrals                              | 139656        |

```

2026-04-01    161453.236770
2026-05-01    161693.434213
2026-06-01    161658.398933
Freq: MS, Name: predicted_mean, dtype: float64

```



*NHS Talking Therapies (IAPT) Referral Forecast for Q1 2026/27. The model (ARIMA 1,1,0) predicts sustained high demand following the Q4 surge, with a projected baseline of ~161,000 referrals per month.*

- The Blue Line (Historical):** This shows the actual referrals recorded in the Monthly\_Time\_Series\_March25\_March26.csv.csv file. You can see the characteristic peaks and troughs that the service has managed over the last year.
- The Red Dashed Line (Forecast):** This is the ARIMA model's prediction. It suggests that demand will stabilize at a higher level (around **161,000** referrals) rather than dropping back down.

3. **The Shaded Pink Area (Uncertainty):** In statistical terms, this is the **95% Confidence Interval**. It tells the clinical leads that while we expect 161k referrals, the "real world" number could reasonably be as high as **210,000** or as low as **112,000**.

## 5. Clinical & Operational Implications :

The forecast of **~161,000 monthly referrals** through June 2026 carries several critical implications for frontline service delivery and strategic planning.

### A. Staffing & Capacity Requirements

The forecast indicates that the high demand seen in early 2026 is likely to persist rather than drop off.

- **Assessment Backlogs:** With referrals remaining above 160k, there is an increased risk of assessment "bottlenecks." Services should review their Triage and Initial Assessment capacity to ensure the **6-week access target** is not compromised.
- **Caseload Management:** Sustained high referral volumes often lead to higher caseloads for Psychological Wellbeing Practitioners (PWPs). Monitoring staff burnout and ensuring adequate supervision is vital during this high-demand period.

### B. Access & Waiting Time Targets

The NHS Long Term Plan sets strict targets for IAPT (Talking Therapies).

- **Target M053 (6-Week Wait):** To maintain the **75% target** for starting treatment within 6 weeks, the model suggests that current resource levels are at their limit.
- **Seasonality Awareness:** Although the "January surge" has passed, the model predicts a "plateau" rather than a decline. Services should avoid significant staffing leave or service pauses in Q1 (April–June) to keep wait times stable.

### C. Resource Allocation & Social Prescribing

Given the high demand, the "High-Intensity" vs. "Low-Intensity" therapy split must be managed carefully.

- **Step-Up/Step-Down Care:** If high-intensity therapy capacity is reached, clinicians may need to lean more heavily on digital-first options or supported self-help to ensure patients aren't left on waiting lists without any support.
- **Community Integration:** High referral numbers often correlate with external stressors (as seen in our anxiety/depression data analysis). Strengthening links

with social prescribing and community support can help manage the "mild-to-moderate" demand that doesn't necessarily require formal clinical intervention.

## 6. Final Conclusion :

The analysis of the **IAPT (Talking Therapies) Referral Data** alongside the **Patient Risk Model** provides a comprehensive view of the challenges and opportunities facing the service in the 2026/27 financial year.

### Summary of Findings :

- **Persistent High Demand:** The ARIMA model indicates that the recent surge in referrals is not a temporary spike but a transition to a higher baseline of approximately **161,000 referrals per month**.
- **Predictable Risk Drivers:** Our patient-level analysis confirms that lifestyle factors—specifically **Sleep Quality** and **Physical Activity**—are the most significant predictors of clinical severity. This suggests that interventions targeting these areas could prevent a portion of referrals from escalating into high-intensity care.
- **Operational Resilience:** While the service is under pressure, the narrow range of the 3-month forecast suggests that demand is currently predictable. This predictability allows for proactive rather than reactive resource management.